Kulkarni, Ryan Booz, and Attila Toth (https://www.timescale.com/blog/time-series-data/). Now, let's switch gears and get practical by looking at several open source and commercial offerings you can use as TSDB backends, especially in the context of Prometheus-compatible cases. We covered the Prometheus basics in chapter 4 and will not repeat them here. We will focus on mid- to large-scale production use cases in this chapter—in other words, when a single Prometheus instance typically is not sufficient anymore. Let's again look at the options for cloud providers, open source offerings, and commercial offerings.

> **NOTE** Prometheus is, by design, not horizontally scalable. That is, you can add more RAM and locally attached persistent storage to scale Prometheus vertically to handle more metrics or deal with a longer time period. What you can't expect is that running multiple copies of Prometheus automatically means the telemetry data is managed for you in such a distributed setup.
>
> If you do have use cases such as a fleet of Kubernetes clusters, where you want to land metrics from many machines in a single place (federation), or you want to be able to query long time periods, such as the past three years, and a single Prometheus instance can't handle the task, you are looking for long-term storage (LTS) solutions. The following options can handle these use cases.

5.3.1 Cloud providers

Every cloud provider has, at the very least, a service in their offerings that allows you to ingest metrics as well as some open source or Prometheus-based ones, in addition to their native offerings. The following are some of the most prominent hyperscaler options, offering both platform and application-metrics ingestions:

- *Amazon CloudWatch Metrics (http://mng.bz/gBne)*—Covers metrics from over 100 AWS services (vended metrics) and keeps them around for 15 months, allowing you to query up-to-the-minute data and historical data equally. You can use the property query language as well as SQL (via CloudWatch Metrics Insights) to query metrics as well as alert (called alarms) on certain conditions. It also comes with dashboards that provide insights out of the box.
- *Amazon Managed Service for Prometheus (http://mng.bz/e169)*—A Prometheus-compatible monitoring service that uses CNCF Cortex in its data plane. You can ingest metrics via the Prometheus remote write interface (http://mng.bz/pPe5) and use PromQL to query it.
- *Amazon Timestream (http://mng.bz/Ox0K)*—A serverless time series database service for IoT and operational applications that offers an SQL-based query language and enables you to use it in a more traditional big data use case.
- *Azure Monitor Metrics (http://mng.bz/Y1Bo)*—Allows you to collect, organize, and analyze metrics from monitored resources in Azure. It supports namespaces to logically group time series as well as rich exploration and fine-grained storage and retention options.